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Ab-Initio Calculations and Thermodynamic Description of the Yb-Cd and Yb-Sn Systems

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Abstract The phase diagrams of Yb-Cd and Yb-Sn systems were calculated by coupling the CALPHAD method and ab initio calculations. The enthalpies of formation of nine binary compounds (YbCd, $\alpha YbCd_2$, Yb_2Sn, αYb_5Sn_3, βYb_5Sn_3, Yb_5Sn_4, YbSn, Yb_3Sn_5 and $YbSn_3$) were determined via ab initio density functional theory using the VASP code. Based on the available experimental data and the computed enthalpies of formation of the compounds, a thermodynamic assessment was carried out. The liquid phases and the γYb(bcc) and βYb(fcc) solid solutions were described by the Redlich–Kister polynomial model, while all the intermetallic compounds were treated as stoichiometric phases. A set of optimized model parameters were obtained for Yb-Cd and Yb-Sn systems. The calculated phase diagrams of these binary systems and their thermodynamic properties are presented and compared with the experimental data.

Keywords ab-initio calculations · CALPHAD · phase diagram · Yb-Cd · Yb-Sn

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1 Introduction

Intermetallic with rare earth alloys have become increasingly used in various fields such as computer memory, thermoelectric, lasers, energy …etc.[1-3] As essential and functional materials, rare earth elements have been named "The Vitamins of Modern Industry". The global demand for these materials is rising every year, this emphasizes the need to improve the existing alloys and to develop new ones.

Phase diagrams play an important role in basic aspects of material research and serve as a guideline for material design. During the last decade, the phase diagrams of various Yb-X systems have been thermodynamically assessed (X = Cu, Ag, Al, Pb, Sb, Ga, Bi, Te, Zn, Ni and Au).[4-13] As a continuation of these studies, The Yb-Cd and Yb-Sn systems are assessed in this paper using CALPHAD method[14,15] and ab initio calculations. The thermodynamic properties and the phase equilibria for the Yb-Cd and Yb-Sn systems would be helpful for a better understanding the interactions of the low melting elements (Cd, Sn) with Ytterbium (Yb). The Yb-Cd system has not been optimized to date while the Yb-Sn system was previously assessed by Idbenali and Servant.[16] In the current work, the Yb-Sn system is reassessed by considering ab initio calculations to supplement the scarcity in the experimental information and then estimate the enthalpy of formation of solid phases to improve the assessment.

2 Literature Review

2.1 Yb-Cd System

The experimental Yb-Cd phase diagram was determined by Palenzona[17] by using differential thermal, microscopic,

and x-ray analysis. The ytterbium used had a purity of 99.9% while the Cadmium had a purity of 99,999%. According to this investigation, six intermetallic compounds were found. Four of them, YbCd, $YbCd_2$,$Yb_{14}Cd_{51}$ and $YbCd_{5.7}$ melt congruently at 1069, 976, 928 and 909 K respectively while the other two, Yb_3Cd_8 and $YbCd_6$, form through peritectic reactions at 927 and 887 K respectively. Five eutectic reactions were obtained at 22.3 at.% of Cd and 740 K, at 61.2 at.% of Cd and 953 K, at 75 at.% of Cd and 921 K, at 82.5 at.% of Cd and 900 K and finally, at about 99.0 at.% of Cd and 590 K. A polymorphic transition of $YbCd_2$ was detected at 968 K. The terminal phases are: γYb (bcc), βYb (fcc) and Cd (hcp) solid solutions; the solid solubility of Yb in (Cd) has not been detected[17,18] while solubility of Cd in (Yb) was determined at ≈1.85 at.% of Cd.

Based on the work of Palenzona,[17] Gschneidner et al.[19] compiled the binary Yb-Cd phase diagram and reported some modifications in the Yb rich side, the eutectic temperature was obtained at 735 K instead of 740 K whereas a significant modification has been made in the catatectic temperature which was 59 K higher than Palenzona's temperature[17]; furthermore, according to Gschneidner et al.,[19] the melting point and the $\gamma \leftrightarrow \beta$ transformation temperatures of pure Yb were at 1092 and 1068 K respectively when Palenzona's diagram[17] showed 1097 and 1009 K for these temperatures.

Crystal structure and lattice parameters of the pure elements and four compounds (YbCd, α$YbCd_2$, $Yb_{14}Cd_{51}$ and $YbCd_6$) are listed in Table 1,[17,20-22] while the structures of the β$YbCd_2$, Yb_3Cd_8 and $YbCd_{5.7}$ intermetallics are unknown.

There are no experimental thermodynamic data for the binary Yb-Cd system reported in the literature up to date due to the considerable loss of Cd (especially above 673 K) and thus difficulty of preparing either the intermetallic compounds or the liquid phase.

2.2 Yb-Sn System

The binary Yb-Sn phase diagram was first established by Palenzona and Cirafici[23] using differential thermal analysis, X ray and micrographic analysis; they reported the existence of five stoichiometric compounds: Yb_2Sn, Yb_5Sn_3, Yb_5Sn_4, YbSn, and $YbSn_3$. The compounds Yb_2Sn and $YbSn_3$ presented a congruent melting, respectively, at 1658 and 1078 K, whereas all the others were formed by peritectic reactions at: 1508 (Yb_5Sn_3), 1433 (Yb_5Sn_4) and 1308 K (YbSn). Three eutectic reactions were found, at 2.0, 64.5 and about 99.5 at.% of Sn and with characteristic temperatures of 1073, 1023 and 503 K respectively. A polymorphic transformation of Yb_5Sn_3 was tentatively assigned at 1363 K. No evidence for mutual solubility was mentioned between Yb and Sn.

Based on their work, Palenzona and Cirafici[24] compiled the Yb-Sn phase diagram and reviewed the Yb terminal side as the purity of the starting Yb was not high (not mentioned by the authors). The Yb was purified with respect to hydrogen in their laboratory by outgassing the commercial product under high dynamic vacuum in a welded thin tantalum crucible. The Yb metal obtained melted at 1088 K and exhibited the allotropic transformation at 1043 K, indicating that the starting Yb was not of high purity. The melting point and the eutectic reaction were adjusted by 4 K above, from 1088 to 1092 and 1073 to 1077 K respectively, while the Yb transformation was adjusted by 25 K, 1068 instead of 1043 K. Later Manfrinetti et al.[25] revised the Yb-Sn phase diagram in the composition range 55-70 at.% of Sn, by means of differential thermal analysis, x ray diffraction, optical and electronic microscopy, a new phase Yb_3Sn_5 was found.

Crystal structure and lattice parameters of the Yb-Sn compounds are listed in Table 1. All binary compounds were examined by Palenzona and Cirafici[23] and Manfrinetti et al.[25] Tsukevaet al.[26] and Wolcyrz et al.[27] have determined the structure of the two varieties of the pure element Sn (α and β).

The enthalpy of formation of $YbSn_3$ was measured by Colinetet al.[28] using a liquid–metal solution calorimetry and by Palenzona[29] using a dynamic differential calorimetric method. Witusiewicz et al.[30] determined by an empirical evaluation, the enthalpies of formation for the compounds $YbSn_3$, Yb_5Sn_3 and Yb_5Sn_4.

The system was optimized by Idbenali and Servant[16] using CALPHAD method, a set of optimized model parameters were obtained but the thermodynamic parameters for the Yb_3Sn_5 phase were not reported.

To clarify the evaluation, an improved reassessment supported by ab initio calculations is carried out in the present work.

3 Ab-Initio Calculations

Due to the scarcity of thermodynamic data, the enthalpies of formation of the intermetallic compounds were determined with ab initio calculations based on density functional theory (DFT),[31] and by using Vienna Ab-initio Simulation Package (VASP).[32,33] Since the structures of β$YbCd_2$, Yb_3Cd_8 and$YbCd_{5.7}$ intermetallics remain undetermined and the unit cells number of atoms was considerable in the $Yb_{14}Cd_{51}$ (68 atoms) and $YbCd_6$ (168 atoms), the calculation was performed only for the YbCd and $YbCd_2$ phases in the Yb-Cd system. The energies of formation at 0 K of the YbCd, $YbCd_2$, Yb_2Sn, Yb_5Sn_3, Yb_5Sn_4, YbSn,Yb_3Sn_5 and $YbSn_3$ compounds are obtained by the following equations:

Table 1 The crystal structure data of the Yb-Cd and Yb-Sn systems

Phase	Comp., at.% Cd or Sn	Pearson symbol and prototype	Space group	Lattice parameters, nm			References
(Yb-β)	0 to ~1.5	cF4 (Cu)	Fm-3 m	0.548	0.548	0.548	[20]
				0.540	0.540	0.540	This work
(Yb-γ)	0 to ~2	*cI2*(W)	Im-3 m	0.387	0.387	0.638	[21]
				0.381	0.381	0.633	This work
YbCd	50	*cP2* (CsCl)	Pm-3 m	0.380	0.380	0.380	[17]
				0.378	0.378	0.378	This work
$YbCd_2$	66.7	hP12 ($MgZn_2$)	$P6_3$/mmc	0.598	0.598	0.959	[17]
				0.603	0.603	0.959	This work
Yb_3Cd_8	72.7	…	…	…	…	…	
$Yb_{14}Cd_{51}$	78.5	hP65($Ag_{14}Gd_{51}$)	P6/m	1.344	1.344	0.965	[17]
$YbCd_{5.7}$	85.1	…	…	…	…	…	
$YbCd_6$	85.7	cI168(Gd_6Y)	Im-3	1.563	1.563	1.563	[17]
(Cd)	~100	hP2(Mg)	$P6_3$/mmc	0.297	0.297	0.561	[22]
				0.301	0.301	0.580	This work
Yb_2Sn	33.3	hP6(Ni_2In)	P63/mmc	0.537	0.537	0.706	[23]
				0.536	0.536	0.705	This work
αYb_5Sn_3	37.5	hP16(Mn_5Si_3)	P63/mcm	0.947	0.947	0.690	[23]
				0.935	0.935	0.672	This work
βYb_5Sn_3	37.5	tI32(Cr_5B_3)	I4/mcm	0.793	0.793	1.468	[23]
				0.810	0.810	1.524	This work
Yb_5Sn_4	44.4	oP36(Gd_5Si_4)	Pnma	0.782	1.581	0.828	[23]
				0.799	1.631	0.841	This work
YbSn	50	tP2(AuCu)	P4/mmm	0.350	0.350	0.440	[23]
				0.346	0.346	0.452	This work
Yb_3Sn_5	0.625	oS32(Pu_3Pd_5)	Cmcm	1.019	0.816	1.037	[23]
				1.029	0.822	1.046	This work
$YbSn_3$	75	cP4($AuCu_3$)	Pm-3 m	0.468	0.468	0.468	[23]
				0.473	0.473	0.473	This work
(Sn-α)	100	cF8(C)	Fd-3 m	0.543	0.543	0.543	[26]
				0.542	0.542	0.542	This work
(Sn-β)	100	tI4(Sn-β)	$I4_1$/amd	0.583	0.583	0.318	[27]
				0.594	0.549	0.321	This work

$$\Delta E\left(Yb_pA_q\right) = E\left(Yb_pA_q\right) - \frac{p}{p+q}E(Yb) - \frac{q}{p+q}E(A) \quad \text{(Eq 1)}$$

where $E(Yb_pA_q)$, E(Yb) and $E(A)$ are the total energies for the Yb_pA_q compound, for pure Yb and for pure A (A = Cd or Sn) respectively. At 0 K, the energy of formation corresponds also to the enthalpy of formation.[34]

The calculations are performed using the project augmented wave (PAW) pseudo-potentials[35,36] with the generalized gradient approximation as implemented by Perdew et al.[37] (GGA-PBE). A plane wave cutoff energy of 500 eV is used for all compounds. *k*-point meshes are compiled using the fully automatic scheme.[38] All calculations are performed using the "Accurate" setting within VASP, and all degrees of freedom of the crystal structures are allowed to relax, including cell shape, volume and atom positions with a preconditioned conjugated gradient (CG) algorithm with the default VASP's convergence criteria.

4 Thermodynamic Models

We describe the Gibbs energy of each phase appearing in the Yb-Cd and Yb-Sn binary phase diagrams including the pure elements Yb, Cd and Sn over an appropriate range of

Table 2 The standard enthalpies of formation of intermetallic compounds in Yb–Cd and Yb-Sn systems

Compound	$\Delta_f H$, kJ/mol	Method	References
YbCd	−42.451	Ab-initio	This work
	−42.021	Calphad	This work
$\alpha YbCd_2$	−36.684	Ab-initio	This work
	−36.196	Calphad	This work
$\beta YbCd_2$	−35.580	Calphad	This work
Yb_3Cd_8	−33.200	Calphad	This work
$Yb_{14}Cd_{51}$	−27.850	Calphad	This work
$YbCd_{5.7}$	−21.603	Calphad	This work
$YbCd_6$	−20.973	Calphad	This work
Yb_2Sn	−63.299	Ab-initio	This work
	−62.898	Calphad	This work, Opt. 1
	−63.305	Calphad	This work, Opt. 2
	−57.560	Calphad	[16]
αYb_5Sn_3	−58.549	Ab-initio	This work
	−62.390	Calphad	This work, Opt. 1
	−63.590	Calphad	This work, Opt. 2
	−58.400	Calphad	[16]
	−53.3 ± 4.4	Vs. N	[30]
	−60.3 ± 5.9	Vs. r_R^{-3}	[30]
β Yb_5Sn_3	−62.285	Ab-initio	This work
	−58.697	Calphad	This work, Opt 1, 2
	−57.602	Calphad	[16]
Yb_5Sn_4	−61.147	Ab-initio	This work
	−61.621	Calphad	This work, Opt. 1
	−63.721	Calphad	This work, Opt. 2
	−56.787	Calphad	[16]
	−58.9 ± 5.2	Vs. N	[30]
	−65.5 ± 7.3	Vs. r_R^{-3}	[30]
YbSn	−64.074	Ab-initio	This work
	−63.990	Calphad	This work, Opt. 1, 2
	−55.077	Calphad	[16]
Yb_3Sn_5	−53.926	Ab-initio	This work
	−53.820	Calphad	This work, Opt. 1, 2
$YbSn_3$	−42.219	Ab-initio	This work
	−42.200	Calphad	This work, Opt. 1, 2
	−41.797	Calphad	[16]
	−44.066*	Experimental	[28]
	−48.002**	Experimental	[29]
	−37 ± 3.8	Vs. N	[30]
	−41.6 ± 8.5	Vs. r_R^{-3}	[30]

Reference state pure Yb(fcc), Cd(hcp) and Sn(bct)

* at 294 K

** at 298 K

Opt. 1 Optimization with VASP data

Opt. 2 Optimization without VASP data

composition and temperature by an adequate model that translates the interactions involved.

4.1 Pure Elements

In the case of the pure element i (i = Yb, Cd, Sn) with φ phase (φ = liquid, βYb(fcc), γYb(bcc), Cd(hcp) or Sn(bct)), the Gibbs energy is described as a function of temperature by the following equation:

$$^0G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{\mathrm{SER}} \quad \text{(Eq 2)}$$

$$^0G_i^{\varphi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad \text{(Eq 3)}$$

where H_i^{SER} is the molar enthalpy of the element i at 298,15 K and 1 bar in its stable element reference (SER) state, $G_i^{\varphi}(T)$ and $^0G_i^{\varphi}(T)$ are the absolute and the relative Gibbs energy of the element i in the φ phase and T is the absolute temperature. The values of the coefficients a to h are taken from the Scientific Group Thermodata (SGTE) compilation by Dinsdale.[39]

4.2 Solution Phases

The liquids and the βYb(fcc) and γYb(bcc) solid solutions were described as disordered solution phases, The Gibbs energy is given as follows:

$$G^{\varphi} = x_{\mathrm{Yb}}^0 G_{\mathrm{Yb}}^{\varphi} + x_i^0 G_i^{\varphi} + \mathrm{RT}[x_{\mathrm{Yb}}\ln x_{\mathrm{Yb}} + x_i \ln x_i] + {}^{ex}G^{\varphi} \quad \text{(Eq 4)}$$

where x_{Yb} and x_i are the mole fractions of the components Yb and i (i = Cd, Sn) respectively, R is the gas constant and φ denotes the phase (φ = liquid, βYb(fcc), γYb(bcc)). $^0G_{\mathrm{Yb}}^{\varphi}$ and $^0G_i^{\varphi}$ are the molar Gibbs energies of pure elements Yb and i in the φ phase respectively, while the term $^{\mathrm{ex}}G^{\varphi}$ is the excess Gibbs energy represented by Redlich–Kister polynomial model[40]:

$$^{\mathrm{ex}}G^{\varphi} = x_{\mathrm{Yb}}x_i \sum_{n=0}^{n=m} {}^nL_{\mathrm{Yb},i}^{\varphi}(x_{\mathrm{Yb}} - x_i)^n \quad \text{(Eq 5)}$$

where $^nL_{\mathrm{Yb},i}^{\varphi}(n = 0, 1, 2, \ldots)$ are the interaction parameters with temperature dependent, written as

$$^nL_{\mathrm{Yb},i}^{\varphi} = a_n + b_nT + c_nT\ln(T) \quad \text{(Eq 6)}$$

a_n, b_n and c_n are parameters to be evaluated.

Table 3 The optimized parameters of Yb-Cd system

Phase	Parameters
Liquid	${}^{0}L^{Liq} = -117026 + 11.247T$
	${}^{1}L^{Liq} = 39068 - 2.102T$
γYb(bcc)	${}^{0}L^{bcc} = -62000$
βYb(fcc)	${}^{0}L^{fcc} = -10000 + 10T$
	${}^{1}L^{fcc} = -56500$
YbCd	$G^{YbCd} = -42021 + 6.759T + 0.5\ G^{SER}_{Yb} + 0.5\ G^{SER}_{Cd}$
$YbCd_2$	$G^{YbCd_2} = -36196 + 2.479T + 0.333\ G^{SER}_{Yb} + 0.667\ G^{SER}_{Cd}$
$YbCd_2$	$G^{YbCd_2} = -35580 + 1.842T + 0.333\ G^{SER}_{Yb} + 0.667\ G^{SER}_{Cd}$
Yb_3Cd_8	$G^{Yb_3Cd_8} = -33200 + 2.225T + 0.273\ G^{SER}_{Yb} + 0.727\ G^{SER}_{Cd}$
$Yb_{14}Cd_{51}$	$G^{Yb_{14}Cd_{51}} = -27850 - 0.18T + 0.215\ G^{SER}_{Yb} + 0.785\ G^{SER}_{Cd}$
$YbCd_{5.7}$	$G^{YbCd_{5.7}} = -21603 - 1.092T + 0.149\ G^{SER}_{Yb} + 0.851\ G^{SER}_{Cd}$
$YbCd_6$	$G^{YbCd_6} = -20973 - 1.0782T + 0.143\ G^{SER}_{Yb} + 0.857\ G^{SER}_{Cd}$

Gibbs energy in J/mol, Temperature (T) in Kelvin

4.3 Stoichiometric Phases

All the intermetallic compounds either in Yb-Cd system: YbCd, $YbCd_2$, Yb_3Cd_8, $Yb_{14}Cd_{51}$, $YbCd_{5.7}$, $YbCd_6$ or in Yb-Sn system: Yb_2Sn, Yb_5Sn_3 (α and β), Yb_5Sn_4, YbSn, Yb_3Sn_5 and $YbSn_3$, were treated as stoichiometric phases. The Gibbs energy in this case is expressed as:

$$G^{\varphi} = x^0_{Yb} G^{fcc}_{Yb} + x^0_i G^{\varphi_i}_i + \Delta_f G^{\varphi} \quad \text{(Eq 7)}$$

With

$$\Delta_f G^{\varphi} = \Delta_f H^{\varphi} - T\Delta_f S^{\varphi} \quad \text{(Eq 8)}$$

where ${}^0G^{fcc}_{Yb}$ and ${}^0G^{\varphi_i}_i$ are the molar gibbs energies of pure elements Yb and i in their standard state respectively and $\Delta_f G^{\varphi}$ is the Gibbs energy of formation of the stoichiometric compound φ, the parameters, $\Delta_f H^{\varphi}$ (enthalpy of formation) and $\Delta_f S^{\varphi}$ (entropy of formation), are obtained by optimization of the available experimental data.

5 Assessment Procedure

Experimental data cited above and calculated enthalpies of formation of the binary compounds were used in the thermodynamic evaluation of all phases in the Yb-Cd and

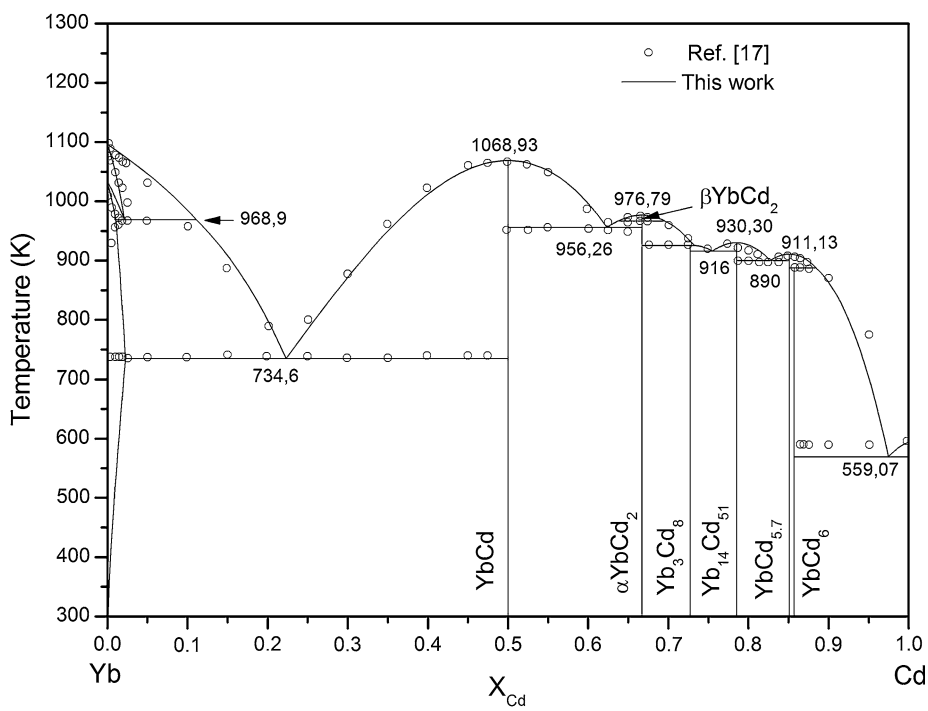

Fig. 1 Comparison between calculated and experimentally measured phase diagrams of the Yb-Cd system

Table 4 The invariant reactions in Yb-Cd system including calculated and experimental results

Reaction	Type of reaction	Compositions of the respective phases, x_{Cd}			T, K	References
Liquid ↔ YbCd	Congruent	0.5	0.5		1068.93	This work
		0.5	0.5		1069 ± 4	[17]
Liquid ↔ $YbCd_2$	Congruent	0.667	0.667		976.79	This work
		0.667	0.667		976 ± 4	[17]
Liquid ↔ $Yb_{14}Cd_{51}$	Congruent	0.785	0.785		930.30	This work
		0.785	0.785		928 ± 4	[17]
Liquid ↔ $YbCd_{5,7}$	Congruent	0.851	0.851		911.13	This work
		0.851	0.851		909 ± 4	[17]
Liquid ↔ YbCd + $\alpha YbCd_2$	Eutectic	0.623	0.5	0.667	956.26	This work
		0.612	0.5	0.667	953 ± 4	[17]
Liquid ↔ Yb_3Cd_8 + $Yb_{14}Cd_{51}$	Eutectic	0.755	0.727	0.785	916.16	This work
		0.75	0.727	0.785	921 ± 4	[17]
Liquid ↔ $Yb_{14}Cd_{51}$ + $YbCd_{5,7}$	Eutectic	0.827	0.785	0.851	899.93	This work
		0.825	0.785	0.851	900 ± 4	[17]
Liquid ↔ (βYb) + YbCd	Eutectic	0.223	0.023	0.5	734.60	This work
		0.223	≈ 0	0.5	740 ± 4	[17]
		0.223	…	0.5	735	[19]
Liquid ↔ $YbCd_6$ + (Cd)	Eutectic	0.975	0.857	1	569.07	This work
		0.99	0.857	1	590 ± 4	[17]
Liquid + $\beta YbCd_2$ ↔ Yb_3Cd_8	Peritectic	0.732	0.667	0.727	925.92	This work
		≈ 0.738	0.667	0.727	927 ± 4	[17]
Liquid + $YbCd_{5,7}$ ↔ $YbCd_6$	Peritectic	0.885	0.851	0.857	887.60	This work
		≈ 0.885	0.851	0.857	887 ± 4	[17]
(γYb) ↔ (βYb) + liquide	Catatectic	0.021	0.010	0.111	968.90	This work
		≈ 0.0318	≈ 0.019	≈ 0.094	969 ± 4	[17]
		…	…	…	1028	[19]
$\beta YbCd_2$ ↔ $\alpha YbCd_2$	Allotropic	0.667	0.667		967.03	This work
		0.667	0.667		968 ± 4	[17]

Yb-Sn systems. The optimization was performed using a computer program developed by Djaballah.[41] A set of optimized model parameters are obtained, each experimental datum is weighted carefully according to its merits, and changed by trial and error during the assessment, reaching a good agreement with the experimental data within the expected uncertainty limits.

6 Results and Discussion

6.1 Ab-Initio Calculations

Lattice parameters and enthalpies of formation for Yb-Cd and Yb-Sn compounds, obtained from ab initio calculations are reported in Tables 1 and 2. The calculated lattice parameters are in good agreement with experimental data, with a maximum difference about 1.9%.

6.2 Thermodynamic Assessment

6.2.1 Yb-Cd System

The best optimization of the Yb-Cd system was obtained with two excess terms for the liquid phase, one for the γYb(bcc) solid solution phase and two for the βYb(fcc). These parameters and those defining the Gibbs energies of formation of the six stoichiometric compounds are reported in Table 3.

The calculated Yb-Cd phase diagram in relation to the experimental phase diagram data of Palenzona[17] is represented in Fig. 1. This Figure shows a good agreement

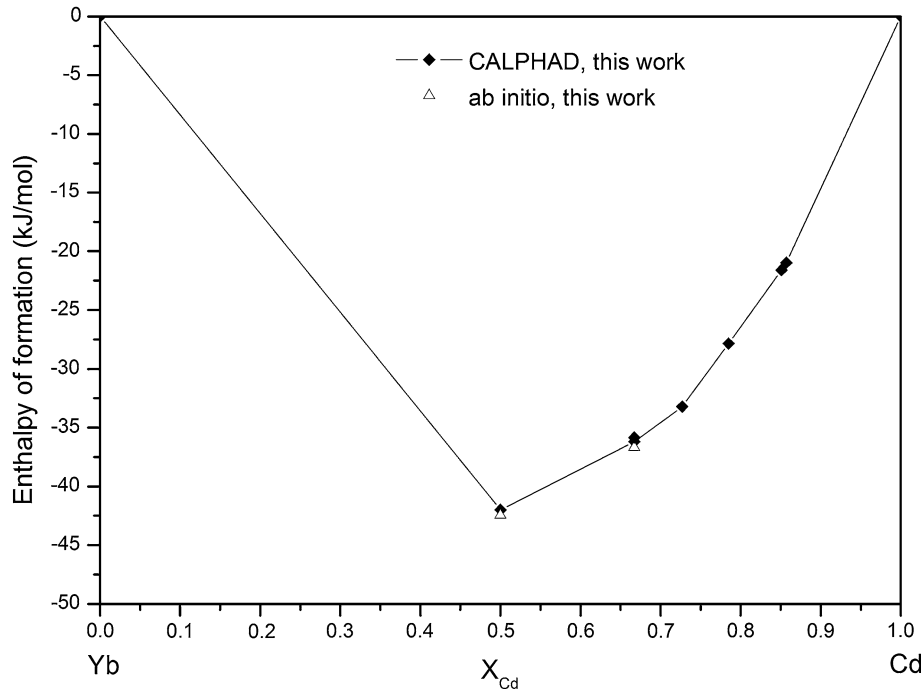

Fig. 2 Standard enthalpies of formation of intermetallic phases in Yb-Cd system. Reference state Yb(fcc) and Cd(hcp)

Table 5 The optimized parameters of Yb-Sn system

Phase	Parameters	Optimization
Liquid	${}^{0}L^{Liq} = -164000 - 7.5T - 0.2TLn(T)$	1,2
	${}^{1}L^{Liq} = 10000 - 6T$	
	${}^{2}L^{Liq} = 40000$	
Yb_2Sn	$G^{Yb_2Sn} = -62898 + 3.879T + 0.667\ G^{SER}_{Yb} + 0.333\ G^{SER}_{Sn}$	1
	$G^{Yb_2Sn} = -63305 + 4.12T + 0.667\ G^{SER}_{Yb} + 0.333\ G^{SER}_{Sn}$	2
Yb_5Sn_3	$G^{Yb_5Sn_3} = -62390 + 2.88T + 0.625\ G^{SER}_{Yb} + 0.375\ G^{SER}_{Sn}$	1
	$G^{Yb_5Sn_3} = -63590 + 3.7324T + 0.625\ G^{SER}_{Yb} + 0.375\ G^{SER}_{Sn}$	2
Yb_5Sn_3	$G^{Yb_5Sn_3} = -58697 + 0.18T + 0.625\ G^{SER}_{Yb} + 0.375\ G^{SER}_{Sn}$	1
	$G^{Yb_5Sn_3} = -58697 + 0.16T + 0.625\ G^{SER}_{Yb} + 0.375\ G^{SER}_{Sn}$	2
Yb_5Sn_4	$G^{Yb_5Sn_4} = -61621 + 1.38T + 0.556\ G^{SER}_{Yb} + 0.444\ G^{SER}_{Sn}$	1
	$G^{Yb_5Sn_4} = -63721 + 2.85T + 0.556\ G^{SER}_{Yb} + 0.444\ G^{SER}_{Sn}$	2
YbSn	$G^{YbSn} = -63990 + 3.90T + 0.5\ G^{SER}_{Yb} + 0.5\ G^{SER}_{Sn}$	1,2
Yb_3Sn_5	$G^{Yb_3Sn_5} = -53820 + 2.72T + 0.375\ G^{SER}_{Yb} + 0.625\ G^{SER}_{Sn}$	1,2
$YbSn_3$	$G^{YbSn_3} = -42200 + 0.33T + 0.25\ G^{SER}_{Yb} + 0.75\ G^{SER}_{Sn}$	1,2

Optimization 1 with VASP data

Optimization 2 without VASP data

Gibbs energy in J/mol, temperature (T) in Kelvin

between the liquidus of the two phase diagrams. A comparison between the experimental and calculated invariant equilibria is presented in Table 4.

All calculated temperatures and compositions of the reactions are well reproduced within the experimental uncertainties except for the eutectic reaction: Liquid → $YbCd_6$ + (Cd), the calculated temperature (569.07 K) was lower than the experimental one (590 K). Also, the eutectic temperature of the invariant reaction: Liquid → (βYb) + YbCd determined in this work at 734.6 K, matches well with the value reported by Gschneidner et al.[19] (735 K) than of Palenzona[17]

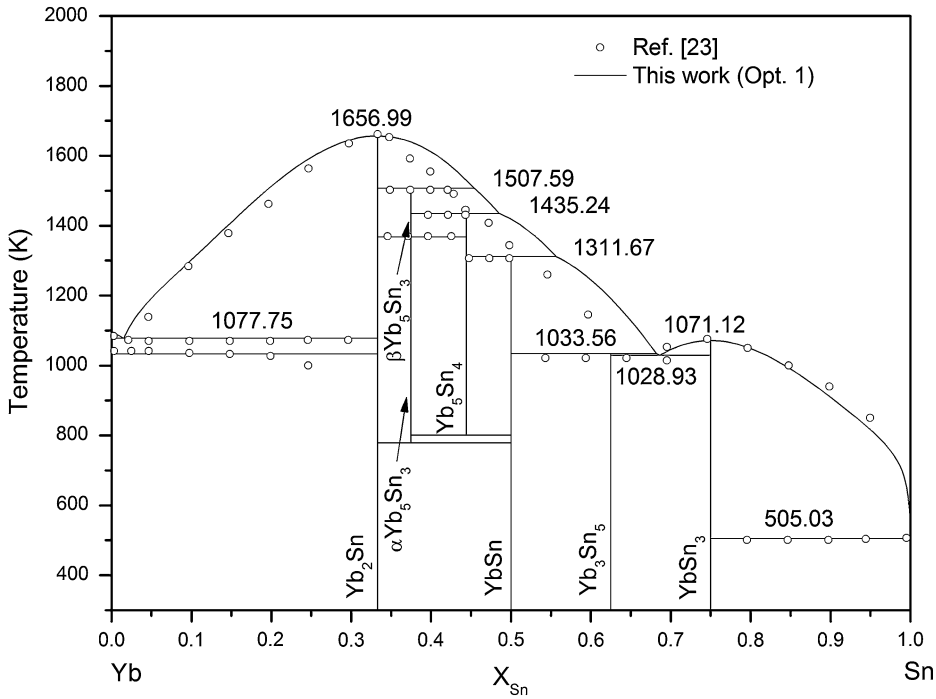

Fig. 3 Comparison between calculated and experimentally measured phase diagrams of the Yb-Sn system (Opt. 1: With VASP data)

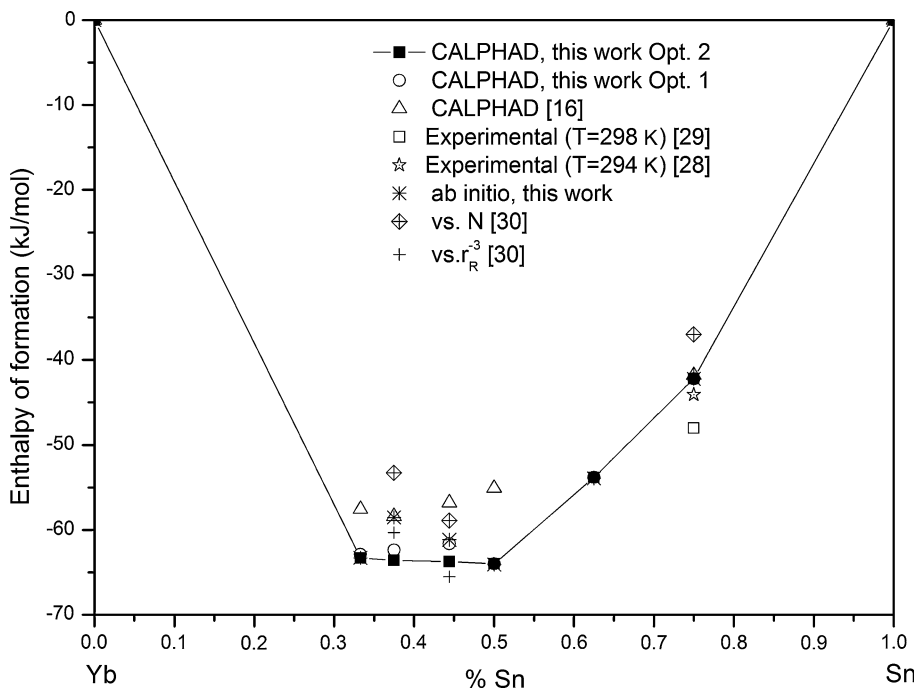

Fig. 4 Standard enthalpies of formation of intermetallic phases in Yb-Sn system. Reference state Yb(fcc) and Cd(hcp) (Opt. 1: With VASP data. Opt. 2: Without VASP data)

(740 K), whereas the catatectic temperature determined at 968.9 K of the invariant reaction: (γYb) → (βYb) + liquid agrees well with the temperature obtained by Palenzona[17] (969 K) than of Gschneidner et al.[19] (1028 K). The calculated eutectic reaction: Liquid → Yb_3Cd_8 + $Yb_{14}Cd_{51}$ is found 1 K out the experimental uncertainties (at 916.16 K) when the experimental one occurs at 921 ± 4 K. Optimized enthalpies of formation found using CALPHAD method with the computed ones determined by ab initio calculations are reported in Fig. 2 (the reference state of Yb and Cd are Yb(fcc) and Cd(hcp) respectively). As shown in Fig. 2, the values obtained are very close for the YbCd and $\alpha YbCd_2$.

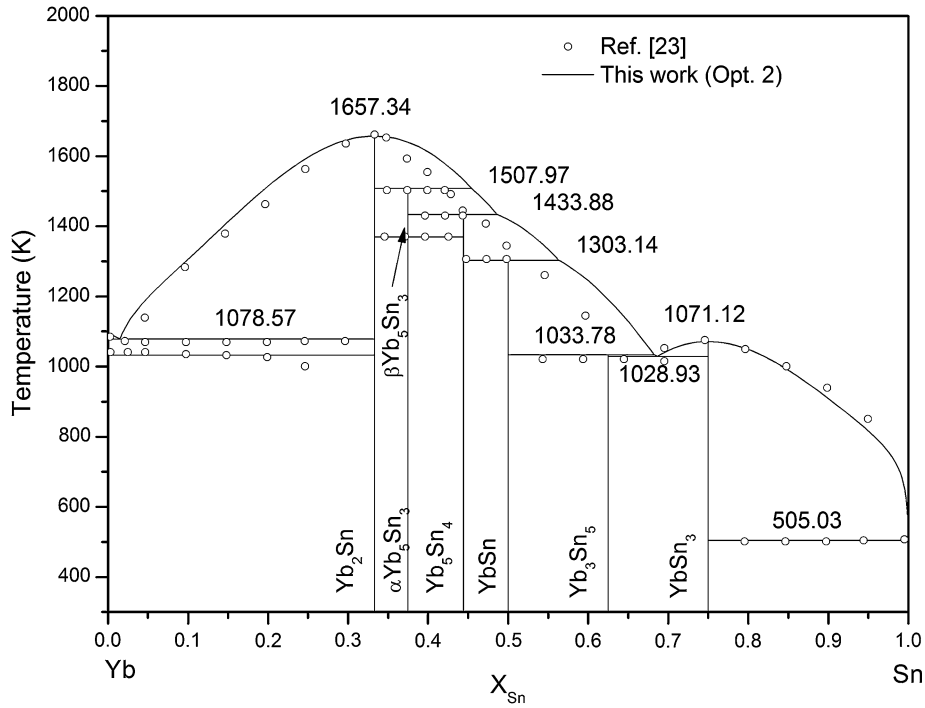

Fig. 5 Comparison between calculated and experimentally measured phase diagrams of the Yb-Sn system (Opt. 2: Without VASP data)

6.2.2 *Yb-Sn System*

Two thermodynamic optimizations were carried out in this system, the values of optimized model parameters are listed in Table 5. In the first optimization we have used the experimental phase diagram data of Palenzona and Cirafici[23] and the enthalpies of formation of all compounds computed by ab initio method.

It has been found that the αYb_5Sn_3 and Yb_5Sn_4 phases are not stable at low temperatures. Figure 3 shows the calculated phase diagram plotted with experimental phase diagram data of Palenzona and Cirafici.[23] The αYb_5Sn_3 compound appears at 779.3 K, whereas the Yb_5Sn_4 compound appears only at 800.86 K. This confirms the ab initio calculations results where these two compounds are not in the ground state line as we can see in Fig. 4.

To improve the assessment, a second optimization was done. We have fixed the excess Gibbs energy parameters of the liquid phase and the formation Gibbs energy of YbSn, Yb_3Sn_5 and $YbSn_3$ compounds, calculated in the first optimization. The enthalpies of formations of the compounds αYb_5Sn_3 and Yb_5Sn_4 are lowered compared to the first optimization (Table 5) and they are stable at room temperature (Fig. 5).

A satisfactory agreement is noted between the calculated phase diagram and the experimental one. The experimental data of the liquidus line in the composition range 33.33% −70 at.% of Sn are slightly lower than the calculated values in the current work; it was difficult to reproduce the liquidus line in this range of composition. Additional experimental phase diagram data are needed in particular on either side of the Yb_3Sn_5 phase which was detected by Manfrinetti et al.[25] The liquidus on the Sn rich side calculated in the present work is in best agreement with the experimental data than the one calculated by Idbenali and Servant.[16]

A comparison between the experimental and calculated invariant equilibria is presented in Table 6. A good agreement between the invariant reaction temperatures of the present calculated and the experimental ones is obtained especially for the congruent reaction: Liquid ↔ Yb_2Sn, the eutectics: Liquid ↔ Yb_2Sn + γYb(bcc) and Liquid ↔ $YbSn_3$ + Sn(bct) and the peritectics: Liquid + Yb_2Sn ↔ βYb_5Sn_3, Liquid + βYb_5Sn_3↔Yb_5Sn_4 and Liquid + YbSn ↔ Yb_3Sn_5.

7 Conclusion

The Yb-Cd system has been thermodynamically assessed for the first time in this work and the Yb-Sn system has been reassessed using CALPHAD method and ab initio approach. As the experimental data are very limited mainly for the Yb-Cd system, ab initio calculations were performed to determine the enthalpies of formation of intermetallic compounds in these systems.

Table 6 The invariant reactions in Yb-Sn system including calculated and experimental results

Reaction	Type of reaction	Compositions of the respective phases, x_{Sn}			T, K	Reference
Liquid → Yb_2Sn	Congruent	0.333			1656.99	This work, Opt. 1
		0.333			1657.34	This work, Opt. 2
		0.333			1658	[23]
		0.333			1663	[16]
Liquid → $YbSn_3$	Congruent	0.75			1071.12	This work, Opt. 1,2
		0.75			1078	[23]
		0.75			1076	[16]
Liquid → Yb_2Sn + γYb(bcc)	Eutectic	0.016	0.333	0	1077.75	This work, Opt. 1
		0.015	0.333	0	1078.57	This work, Opt. 2
		0.02	0.333	0	1077	[24]
		0.02	0.333	0	1074	[16]
Liquid → Yb_3Sn_5 + $YbSn_3$	Eutectic	0.686	0.625	0.75	1028.93	This work, Opt. 1,2
		0.665	0.625	0.75	1023	[25]
		0.645	0.625	0.75	1023	[23]
		0.676	0.625	0.75	1026	[16]
Liquid → $YbSn_3$ + Sn(bct)	Eutectic	0.9998	0.75	1	505.03	This work, Opt. 1,2
		0.9995	0.75	1	503	[23]
		0.9995	0.75	1	505	[16]
Liquid + Yb_2Sn → βYb_5Sn_3	Peritectic	0.454	0.333	0.375	1507.59	This work, Opt. 1
		0.454	0.333	0.375	1507.97	This work, Opt. 2
		≈0.425	0.333	0.375	1508	[23]
		0.457	0.333	0.375	1505	[16]
Liquid + βYb_5Sn_3 → Yb_5Sn_4	Peritectic	0.485	0.375	0.444	1435.24	This work,Opt. 1
		0.486	0.375	0.444	1433.88	This work, Opt. 2
		≈0,451	0.375	0.444	1433	[23]
		0.493	0.375	0.444	1428	[16]
Liquid + Yb_5Sn_4 → YbSn	Peritectic	0.557	0.444	0.5	1311.67	This work, Opt. 1
		0.563	0.444	0.5	1303.14	This work, Opt. 2
		≈0.513	0.444	0.5	1308	[23]
		0.56	0.444	0.5	1311	[16]
Liquid + YbSn → Yb_3Sn_5	Peritectic	0.682	0.5	0.625	1033.56	This work, Opt. 1,2
		≈0.631	0.5	0.625	1033	[25]
		0.67	0.5	0.625	1032	[16]
βYb_5Sn_3 → αYb_5Sn_3	Allotropic	0.375			1367.78	This work, Opt. 1
		0.375			1369.67	This work, Opt. 2
		0.375			1363	[23]
		0.375			1360	[16]

Opt. 1 Optimization with VASP data

Opt. 2 Optimization without VASP data

A set of optimized thermodynamic parameters describing the Gibbs energy of each solution phase and intermetallic compound for the Yb-Cd and Yb-Sn systems have been developed. Calculated phase diagrams from the thermodynamic parameters are in good agreement with the experimental ones.

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